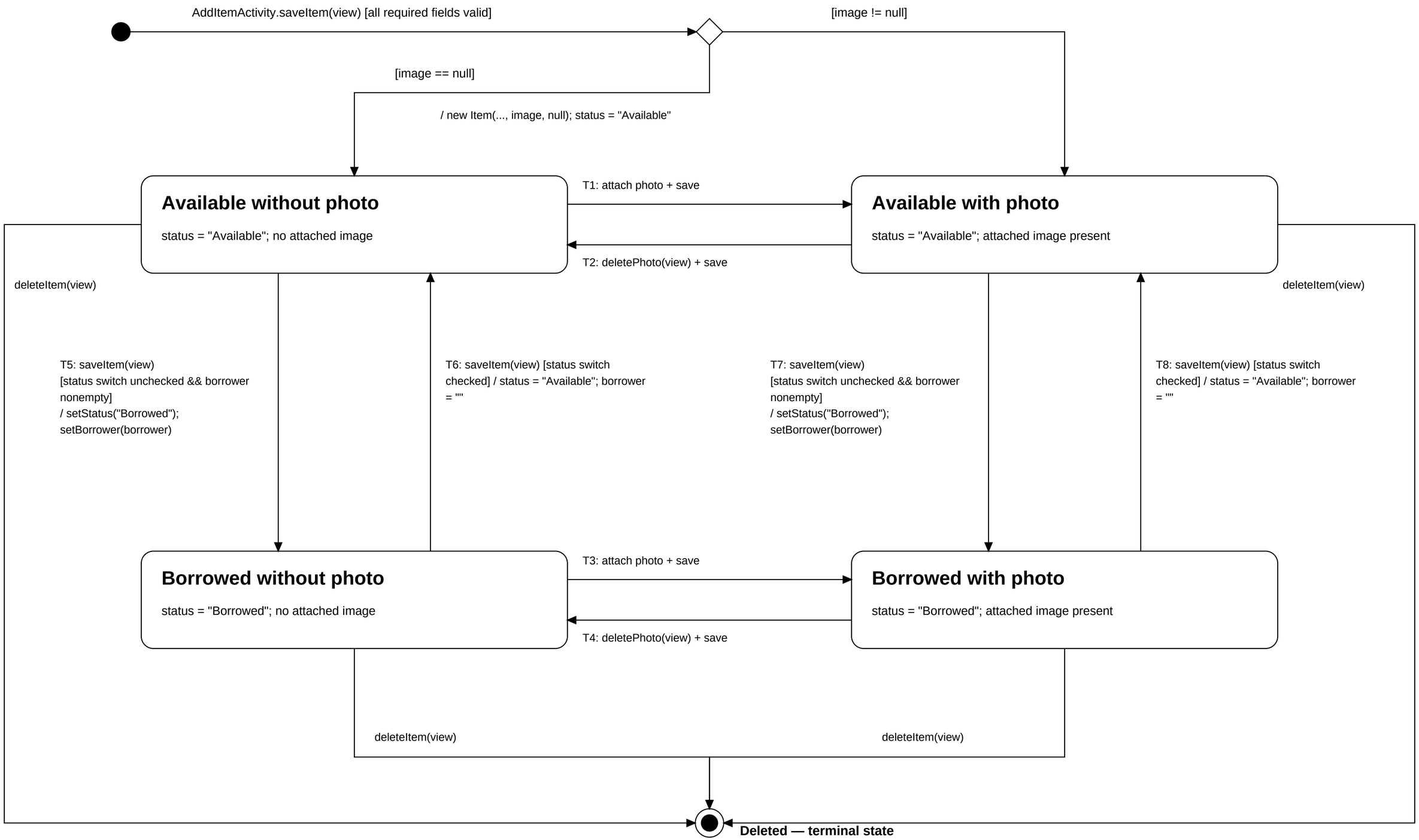
SharingApp — Item Lifecycle | UML State Diagram

Four states combine availability and the presence of an attached photo. The state machine tracks the logical item, identified by its id.



T1: Available without -> Available with; T2: reverse. T3: Borrowed without -> Borrowed with; T4: reverse.

T5: Available without -> Borrowed without; T6: reverse. T7: Available with -> Borrowed with; T8: reverse.

Attach photo means addPhoto(view), then successful onActivityResult(... RESULT_OK ...) with a non-null Bitmap, followed by saveItem(view).

All saving transitions require nonempty title, maker, description, length, width and height. Borrowed also requires a nonempty borrower. Failed validation does not commit a replacement item.

Edit save rebuilds the Item with the same id, replaces it in ItemList and calls saveItems(context). This replacement is not terminal deletion. Explicit delete removes the logical item.

If both the status and photo change in one edit, the target is determined by both final values; all four combined transitions are listed on page 2. Saving unchanged values is a self-transition.

The diagram models logical saved edits. toggleSwitch() can modify the current in-memory item before save; file-write exceptions are caught internally, so durable persistence is not guaranteed.

SharingApp — Transition Details and Source Mapping

The state diagram and this transition dictionary form one model. "save" means a valid EditItemActivity.saveItem(view).

ID	FROM	TO	EVENT / GUARD
T1	Available without photo	Available with photo	Photo attached; status remains Available
T2	Available with photo	Available without photo	Photo deleted; status remains Available
T3	Borrowed without photo	Borrowed with photo	Photo attached; status remains Borrowed
T4	Borrowed with photo	Borrowed without photo	Photo deleted; status remains Borrowed
T5	Available without photo	Borrowed without photo	Unchecked switch; nonempty borrower; no photo
T6	Borrowed without photo	Available without photo	Checked switch; photo remains absent
T7	Available with photo	Borrowed with photo	Unchecked switch; nonempty borrower; photo retained
T8	Borrowed with photo	Available with photo	Checked switch; photo retained
T9	Available without photo	Borrowed with photo	Save with unchecked switch, borrower and attached photo
T10	Borrowed with photo	Available without photo	Save with checked switch and deleted photo
T11	Available with photo	Borrowed without photo	Save with unchecked switch, borrower and deleted photo
T12	Borrowed without photo	Available with photo	Save with checked switch and attached photo

Self-transitions: each of the four states returns to itself when saveItem(view) succeeds without changing status or photo presence. Editing title, maker, description or dimensions, or replacing an existing photo, does not create an additional state.

Final-state mapping: from every state, EditItemActivity.deleteItem(view) invokes item_list.deleteItem(item), then item_list.saveItems(context), and navigates to MainActivity. No outgoing transition from the final state is modeled.

Creation mapping: AddItemActivity.saveItem(view) constructs Dimensions, then Item(..., image, null), adds it to ItemList and saves. Item initializes status to Available and borrower to the empty string. The creation choice therefore reaches one of the two Available states, depending on whether an image is attached.

Editing mapping: onCreate loads the selected item and its image. deletePhoto sets the editor image to null. A successful camera result supplies the editor image. saveItem constructs a replacement with the original id; an unchecked switch sets Borrowed and the supplied borrower, while a checked switch keeps the constructor defaults Available and empty borrower.

Scope: four stable logical states, initial pseudostate, guarded creation choice, all single-axis changes, combined saved edits, self-transitions and explicit final deletion. The transient editor UI and storage-error handling are summarized rather than represented as extra Item states.

Source: supplied SharingApp-c1-items-only-v4 archive; AddItemActivity.java, EditItemActivity.java and Item.java.